

Full Stack Development with MERN: Project Documentation

1. Introduction

- **Project Title:** ShopEZ — Modern MERN Stack E-Commerce Platform.
- **Team Members & Roles:** Venu(Individual) --Lead Full-Stack Developer & Database Architect (Responsible for end-to-end React UI development, Express REST API engineering, and decoupled database design).

2. Project Overview

Purpose

ShopEZ is designed to lower the barrier to digital transformation for small-to-medium retail enterprises (SMEs) by providing a lightweight, self-hosted, high-performance e-commerce software suite. It targets critical industry pain points: reducing the cognitive load caused by bloated administrative panels, eliminating irrelevant interface inputs for varied product lines, and preventing high shopping cart abandonment rates caused by lost session states across modern web browsers.

Features

- **Adaptive Customer Storefront:** A fluid Single Page Application (SPA) that reads product categories dynamically and automatically modifies its interface layout (e.g., hiding apparel sizing dropdowns when displaying Electronic goods).
- **Persistent Shopping State:** Ensures persistent local state synchronization for shopping carts and user wishlists using browser caching mapped directly to global user contexts.
- **Decoupled Admin Portal:** A secure, real-time administrative cockpit providing instant access to business metrics (total revenue, active order volume, total customer profiles).
- **Isolated Invoice Printing:** Built-in print styling leveraging specialized CSS media queries (@media print) to completely isolate invoice details, stripping away web UI layouts for high-fidelity physical bill printing.
- **Role-Based Route Guarding:** Automatically identifies user states via authorization headers and triggers deterministic routing, instantly sending administrators to the metrics dashboard upon logging in.

3. Architecture

Frontend

The frontend layer is engineered using React.js compiled via Vite for maximum build-time performance and optimal static bundling. Global user actions, session variables, and UI toggles are managed through the React Context API via decoupled contexts (AuthContext, CartContext, WishlistContext). Responsive utilities are styled via Tailwind CSS, and all backend server communications are handled asynchronously via Axios client instances.

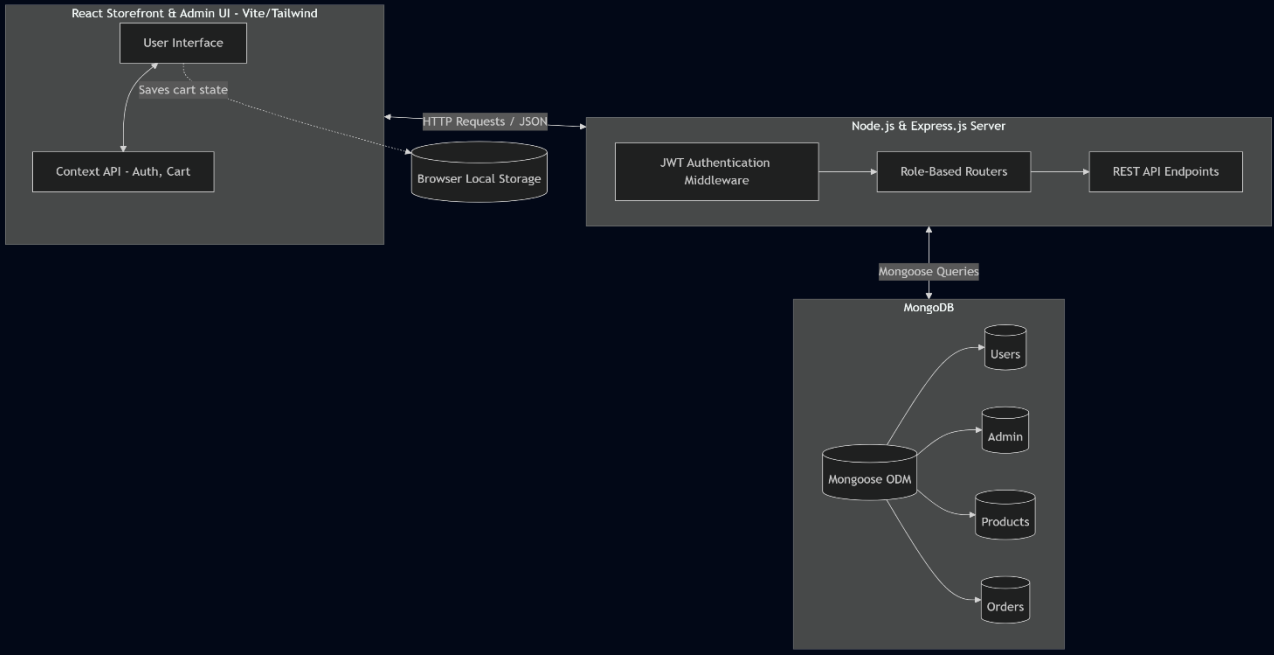
Backend

The backend application layer is built as a stateless Node.js and Express.js RESTful API. It organizes logic into distinct controller frameworks to manage authentication, catalog access, client baskets, and transaction records. Routes are strictly protected using customized JSON Web Token (JWT) verification middleware.

Database

The system utilizes MongoDB Atlas as its cloud database, interacting via the Mongoose Object Data Modeling (ODM) library to enforce structural schemas and handle relationship validation. The database features an isolated architectural design:

- **Standard Customer Collection (users):** Stores customer parameters, including sub-document schema arrays for shippingAddresses and tokenized payment structures (savedPaymentMethods).
- **Decoupled Admin Collection (admins):** A completely isolated collection handling role-based access control (RBAC), fine-grained strings for permissions, brute-force prevention flags (failedAttempts), and an internal sub-document auditLog array. This total separation prevents privilege escalation vectors.



ADMINS			
ObjectId	_id	PK	
String	username		
String	email		
String	department		SuperAdmin, Sales, etc.
Array	permissions		['read:products', etc.]
Object	securityFlags		2FA, failed attempts
Array	auditLog		Tracks admin actions

USERS			
ObjectId	_id	PK	
String	username		
String	email		
String	password		
Boolean	isBlocked		
Array	shippingAddresses		Sub-documents
Array	savedPaymentMethods		Tokenized cards

PRODUCTS			
ObjectId	_id	PK	
String	title		
String	category		
Number	price		
Number	stock		Total inventory
Map	sizeStock		{S:10, M:5}
Array	sizes		

ORDERS			
ObjectId	_id	PK	
ObjectId	userId	FK	Ref: users
String	name		
String	paymentStatus		
String	orderStatus		
Date	orderDate		

ORDER_ITEMS			
ObjectId	productId	FK	Ref: products
String	title		Snapshotted title
String	size		Selected size
Number	quantity		
Number	price		Snapshotted price

places

snapshot referenced in

contains

4. Setup Instructions

Prerequisites

- Node.js (v18.x or higher installed locally)
- npm (v9.x or higher)
- MongoDB Atlas Cloud Account (or Local MongoDB Community Server instance)

Installation

1. Clone the Repository:

Bash

```
git clone https://github.com/venu/shopez-ecommerce.git  
cd shopez-ecommerce
```

2. Install Client Dependencies:

Bash

```
cd client  
npm install
```

3. Install Server Dependencies:

Bash

```
cd ../server  
npm install
```

4. Configure Environment Variables:

Create a .env file inside the root /server directory:

Code snippet

```
PORT=5000
MONGO_URI=mongodb+srv://<username>:<password>@cluster0.mongodb.net/shopezDB
JWT_SECRET=your_super_secret_cryptographic_jwt_key_2026
```

5. Folder Structure

The workspace utilizes a clean repository split separating the client code from backend runtime environments:

Project Structure

```
ShopEZ/
├── client/                                # Frontend React Application
│   ├── src/
│   │   ├── components/                  # Reusable layout & common UI components
│   │   ├── context/                    # React Context providers (Auth, Cart, Wishlist)
│   │   ├── pages/                      # Page views (Admin Dashboard, Checkout, Cart, Products, etc.)
│   │   ├── routes/                    # Protected and public routing definitions
│   │   ├── services/                  # Axios API integrations
│   │   ├── index.css                   # Core Tailwind theme configuration
│   │   └── main.jsx                    # App entry point
│   ├── vite.config.js
│   └── package.json
├── server/                              # Backend REST API Application
│   ├── config/                        # DB connection config
│   ├── controllers/                   # Express request controllers (Auth, Products, Orders, Admin)
│   ├── models/                       # Mongoose schemas (User, Product, Order, Cart)
│   ├── routes/                       # Express route mappings
│   ├── seed.js                       # Initial database seeding script
│   ├── index.js                      # Server entry point
│   └── package.json
```

6. Running the Application

Commands to start both services concurrently during local development workflows:

- **Frontend Execution:**

Bash

```
cd client
npm start
```

- **Backend Execution:**

Backend (/server)

- `npm run start` or `node index.js` : Runs the API server.
- `npm run seed` : Clears existing records and seeds fresh demo catalogs and credentials.

7. API Documentation

All core endpoints exposed by the Node/Express backend infrastructure:

1. User Authentication

- **Method / Route:** POST /api/auth/login
- **Parameters (Request Body JSON):**

JSON

```
{
  "email": "customer@email.com",
  "password": "SecurePassword123"
}
```

Example Response (200 OK): [PDF](#)

JSON

```
{
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "userType": "customer",
  "username": "CustomerUser"
}
```

2. Retrieve Product Catalog

- **Method / Route:** GET /api/products
- **Parameters:** None (Optional query tracking for filters)
- **Example Response (200 OK):**

JSON

```
[
  {
    "_id": "64a7b2e1f3d4c5b6a7f8e9d0",
    "title": "Premium Wireless Headphones",
    "price": 8999,
    "category": "Electronics",
    "sizes": [],
    "stock": 45,
    "sizeStock": {}
  }
]
```

3. Create Transaction Order

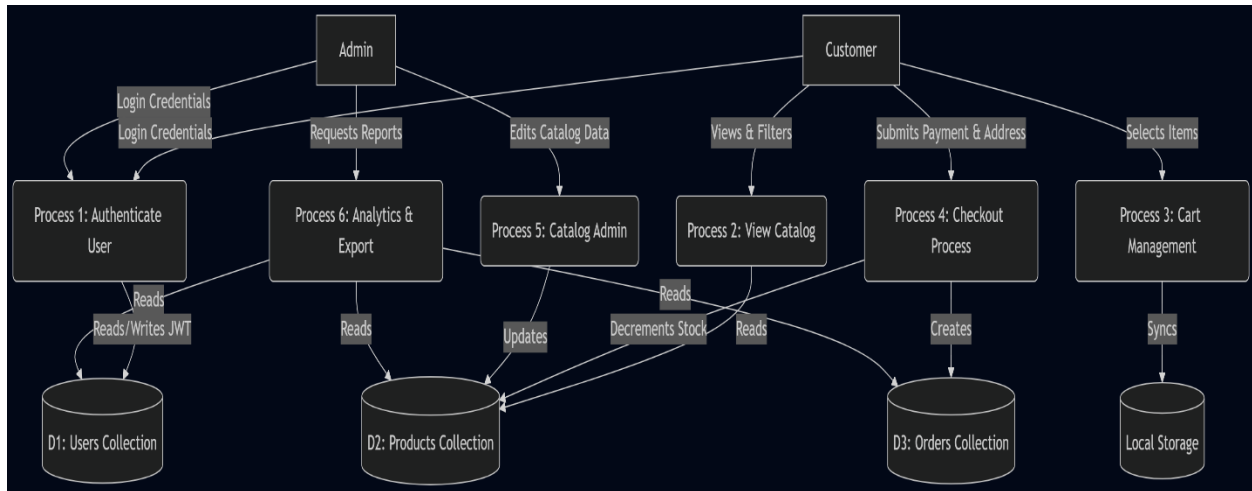
- **Method / Route:** POST /api/orders
- **Parameters (Headers & Request Body JSON):**
 - *Header:* Authorization: Bearer <JWT_TOKEN>
 - *Body:*

JSON

```
{
  "name": "Venu",
  "email": "venu@university.com",
  "mobile": "9876543210",
  "address": "Uppal, Hyderabad, Telangana",
  "pincode": "500039",
  "paymentMethod": "COD",
  "orderItems": [
    {
      "productId": "64a7b2e1f3d4c5b6a7f8e9d0",
      "title": "Premium Wireless Headphones",
      "size": "Oversized",
      "quantity": 1,
      "price": 8999
    }
  ]
}
```

JSON

```
{  
  "message": "Order created successfully",  
  "orderId": "65b9c1d2e3f4a5b6c7d8e9f0"  
}
```



8. Authentication

Authentication and authorization rely on stateless JSON Web Tokens (JWT) combined with industry-standard security components:

- **Password Hashing:** Passwords are cryptographically salted and hashed using bcrypt before being stored in the database.
- **Token Issuance & Lifecycles:** When a client authenticates via login routes, the server signs a token embedding the database `_id` and the user's explicit authorization role (`userType`).

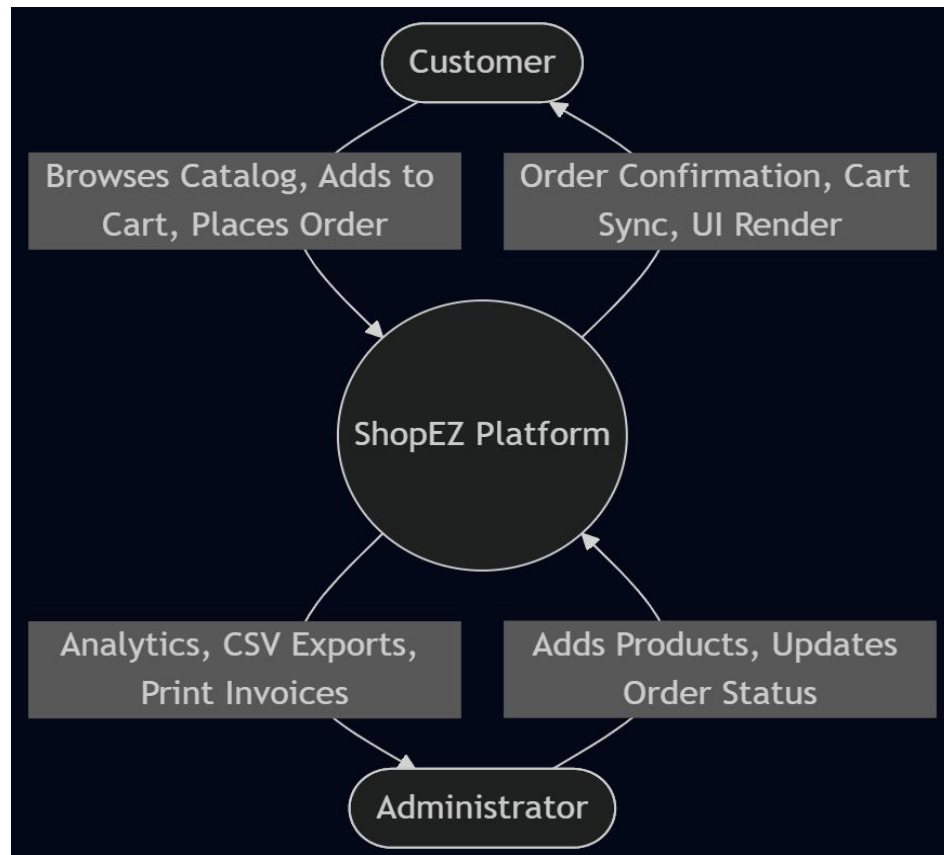
- **Authorization Header Verification:** The client caches this token inside the application's global React context and attaches it to all outgoing secure requests inside the HTTP Authorization: Bearer <token> header. Express routers pass requests through custom middleware to parse headers, confirm signatures, and block unauthorized traffic with structural 403 Forbidden errors.

9. User Interface

The frontend user experience maps to a clean, single-page interface layout:

- **Customer Storefront Homepage:** Renders a clean grid layout presenting available catalog items.
- **Context-Aware Details Module:** Dynamically hides sizing UI matrices for product categories like "Electronics" or "Home Decor", presenting simple inventory metrics.
- **Admin Reporting Dashboard:** A desktop view providing data management controls, table exports, and single-click layout printing.

[Developer Note: Embed file references or upload screenshots/GIFs showing active storefront panels and print-preview mockups here]



10. Testing

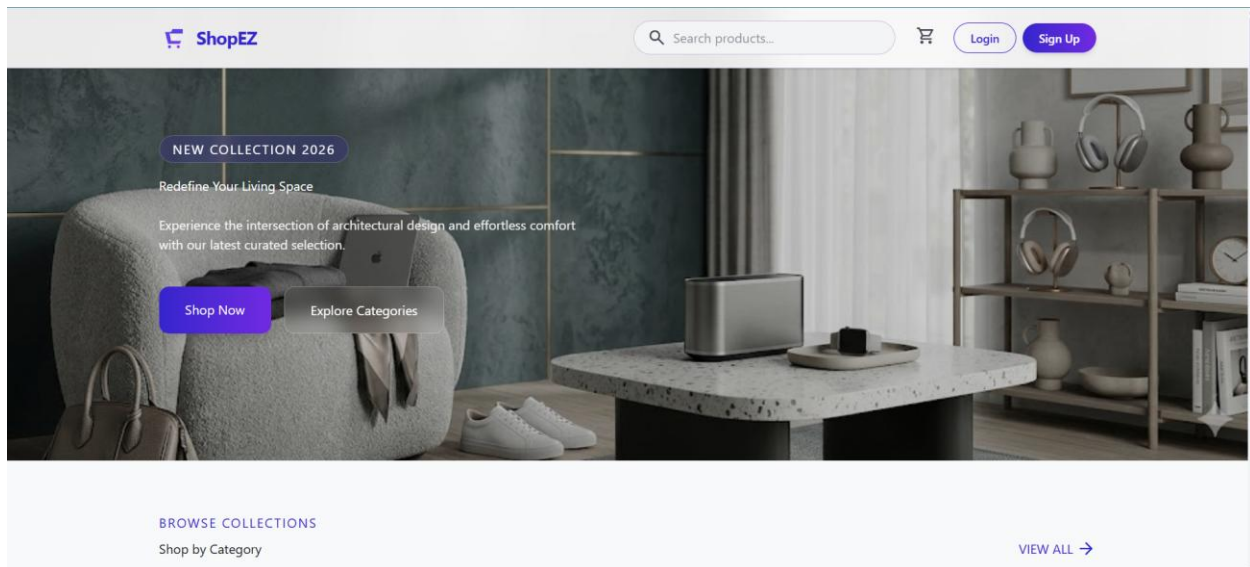
The platform testing strategy focuses on automated interface verification and manual routing validation:

- **Manual User Acceptance Testing (UAT):** Completed structured test case models tracking contextual UI rendering actions, interface layout changes, and programmatic role-based redirects.
- **API Validation Processes:** Endpoint routing behaviors, header injection processing, JSON parsing models, and error responses were fully validated using Postman API execution collections.

11. Screenshots or Demo

- **Staging Storefront URL:** <http://localhost:3001>
- **Staging API Endpoint:** <http://localhost:8000>

[Developer Note: Insert local deployment screenshot links or video walk-through embeds directly here]



ShopEZ

Welcome back

Enter your credentials to access your account.

LOGIN

REGISTER

Email Address

name@example.com

Password

Forgot password?

Remember Me

Sign In

© 2026 ShopEZ Premium E-commerce.
All rights reserved.

Back to Home

ShopEZ

Search products...

♥

🛒

🔒

Filters

Clear All

SEARCH

Search...

GENDER

Men

Women

Kids

Unisex

CATEGORIES

Electronics

Lifestyle

Apparel

Home Decor

Shirts

PRICE RANGE

Premium Collection

Showing 1–9 of 9 items

Sort by: Popularity

NEW

ELECTRONICS

Aura Series Head...

₹299.00

★ 3.1

NEW

LIFESTYLE

Minimalist Horiz...

₹450.00

★ 4.5

-15%

APPAREL

Apex Runner V2

₹220.00

₹187.00

★ 4.2

NEW

ELECTRONICS

Retro Frame X100

₹1199.00

★ 4.7

NEW

NEW

NEW

NEW



Search products...



Shopping Bag
You have 2 items in your bag



Aura Series Headphones
₹299.00 0

—

1

+





Minimalist Horizon Watch
SIZE: M
₹450.00 0

—

1

+



Order Summary

Subtotal (2 items)

₹749.00

Shipping

FREE

Estimated Tax

₹0.00

Order Total

₹749.00

Proceed to Checkout →

Continue Shopping



Search products...



**Shipping**

Payment

Review

Shipping Address

venu yadav
12,Uppal
PIN: 724723

Change

Payment Method

Credit / Debit Card
Secure payment via Stripe

UPI Payment
Pay using GPay, PhonePe, or BHIM

Cash on Delivery

Order Summary



Aura Series Headphones
Qty: 1
₹299.00



Minimalist Horizon Watch
Size: M | Qty: 1
₹450.00

Subtotal

₹749.00

Shipping

Free

Estimated Tax

₹59.92

Total

₹808.92

ShopEZ

Search products...

Order Success

Order History

Order History

Manage and track your recent premium purchases.

All Orders

#EZ-C33532

Placed on 6/17/2026

order placed

₹450.00

Minimalist Horizon Watch

M • Qty 1

Arriving 6/24/2026

Invoice

Buy Again

#EZ-C3352D

Placed on 6/17/2026

order placed

₹299.00

Aura Series Headphones

Qty 1

Arriving 6/24/2026

Invoice

Buy Again

ShopEZ

Search products...

My Wishlist

2 items saved

LIFESTYLE

Minimalist Horizon Watch

₹450.00

Add

ELECTRONICS

Aura Series Headphones

₹299.00

Add

ShopEZ Admin Portal

DashboardProductsOrdersUsersLogout

Welcome back, Admin
June 17, 2026

Export DataAdd Product

+24% ↗

TOTAL REVENUE
₹9,426

+12% ↗

NEW ORDERS
15

+3% ↗

ACTIVE PRODUCTS
9

+12% ↗

TOTAL CUSTOMERS
3

Quick Actions

Manage Products

Manage Orders

Manage Users

Recent Orders

View All Orders

Order ID	Customer	Status	Date	Amount	Actions
----------	----------	--------	------	--------	---------

ShopEZ Admin Portal

DashboardProductsOrdersUsersLogout

Back to Dashboard

Inventory Management
Oversee your catalog and product listings.

Add New Product

Search inventory...

Product	Category	Price	Discount	Pieces	Actions
Aura Series Headphones ID: PROD-E98DB4	Electronics (Unisex)	₹299.00	-	14 pcs	✎🗑
Minimalist Horizon Watch ID: PROD-E98DB5	Lifestyle (Unisex)	₹450.00	-	0 pcs	✎🗑
Apex Runner V2 ID: PROD-E98DB6	Apparel (Men)	₹220.00	15% OFF	0 pcs	✎🗑
Retro Frame X100 ID: PROD-E98DB7	Electronics (Unisex)	₹1199.00	-	0 pcs	✎🗑
Oslo Classic Frames ID: PROD-E98DB8	Apparel (Unisex)	₹210.00	-	0 pcs	✎🗑
Sonic Sphere Hub ID: PROD-E98DB9	Electronics (Unisex)	₹325.00	-	0 pcs	✎🗑

ShopEZ Admin Portal

DashboardProductsOrdersUsersLogout

Back to Dashboard

Manage Orders

Review and update customer transactions and delivery states

ORDER ID	CUSTOMER	PRODUCT NAME	DATE	AMOUNT	STATUS	UPDATE STATUS
#C33532	venu yadav venu@example.com	Minimalist Horizon Watch Qty: 1 Size: M	6/17/2026	₹450.00	ORDER PLACED	Placed
#C3352D	venu yadav venu@example.com	Aura Series Headphones Qty: 1	6/17/2026	₹299.00	ORDER PLACED	Placed
#BA38F7	venu yadav venu@example.com	US Polo Cotton Shirt White Qty: 1 Size: S	6/13/2026	₹2500.00	ORDER PLACED	Placed
#BA3854	venu yadav venu@example.com	Apex Runner V2 Qty: 1 Size: M	6/13/2026	₹187.00	SHIPPED	Shipped
#BA384F	venu yadav venu@example.com	Minimalist Horizon Watch Qty: 1 Size: M	6/13/2026	₹450.00	DELIVERED	Delivered
#BA384A	venu yadav venu@example.com	Origin Espresso Brewer Qty: 1	6/13/2026	₹890.00	CANCELLED	Cancelled
#A8C5C7	venu yadav venu@example.com	Retro Frame X100 Qty: 1 Size: One Size	6/13/2026	₹1199.00	ORDER PLACED	Placed
#458CE3	John Doe customer@shopez.com	Sonic Sphere Hub Qty: 1 Size: M	6/13/2026	₹325.00	DELIVERED	Delivered

ShopEZ Admin Portal

DashboardProductsOrdersUsersLogout

Back to Dashboard

Customer Management

Review registered user accounts and authorization roles.

User	Email Address	Joined Date	Role	Management
T Test User	testuser@example.com	6/13/2026	CUSTOMER	BLOCK USERDELETE
V venu yadav	venu@example.com	6/13/2026	CUSTOMER	BLOCK USERDELETE
J John Doe	customer@shopez.com	6/13/2026	CUSTOMER	UNBLOCKDELETE

Demo Video : [Click Here](#)

12. Known Issues

- **UI Flashing on Slow Connections (BG-001):** The sizing selectors occasionally flash on screen briefly before hiding when loading electronic products under slow cellular connections.
- **Counter Event Queue Lag (BG-002):** Rapidly executing repeated clicks on "Add to Cart" triggers slight event processing lag within the global navbar bubble update count.

13. Future Enhancements

- **Multi-Store Synchronization Engine:** Upgrading the application backend to support segmented multi-tenant stores and localized warehouse configurations from a single deployment.
- **Point-of-Sale (POS) Integrations:** Developing plugin micro-services to sync digital MongoDB database inventories with physical barcode scanners and checkout points.
- **Automated Marketing Suite:** Implementing automated client workflows for email/SMS recovery reminders to re-engage shoppers after cart abandonment triggers.